

The Mirror at $W = 0$

The Detailed Fluctuation Theorem as a Straight Line Through the Origin

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

The thermodynamic uncertainty relation of Ch.76 priced the act of knowing in variance. This chapter opens the DETAILED ledger beneath it: on a time-reversible cyclic protocol with $\Delta F = 0$, the Crooks fluctuation theorem (Crooks 1999; Jarzynski 1997) fixes the ratio of the two arms of the work distribution at every level, $P(+W)/P(-W) = e^{W}$. The "mirror graph" $M(w) = \ln P(w) - \ln P(-w)$ is therefore a STRAIGHT LINE $M(w) = w$ through the origin, slope 1, and the ratio at $w = 0$ is a 0/0 whose removable value is 1. Measured on the same two instruments as Ch.76: on the control round trip the mirror is exactly the line slope 1 (0.996 $\pm$ 0.01 in the wings, where the integral ledger of Ch.71 cannot reach), while the variance ledger still reads $q = 0.894$ - the slack of Ch.76 lives entirely OFF the mirror, in the bending of P itself. The demon's coin leans the mirror: engaged slope -0.25, harvest -5.6, with $\langle e^{-W} \rangle = 1.075$ and 1.138 (the J_{act} ledger of Ch.74/75) - information moves both ledgers together. The Gaussian tangent ($\mu = +5.98$, $\sigma = 3.47$) has no mirror at our resolution: its negative arm is a rare event, and its only ledger is the integral one. Two ledgers, one book: the mirror balances exactly, the variance books the fee, and the removable value is 1 in both.

1. The Detailed Ledger

Jarzynski (PRL 78, 2690, 1997) and Crooks (PRE 60, 2721, 1999) gave fluctuation theorems in two versions - the integral one $\langle e^{-W} \rangle = 1$, an area beneath the distribution, and the DETAILED one, a pointwise mirror. For a protocol $\lambda(t) = \lambda(T-t)$ (time-reversible) and $\Delta F = 0$,

$$P(+W) / P(-W) = e^{\{W\}} \quad \text{for every } W \text{ where both densities exist}$$

$$\text{mirror } M(w) = \ln P(w) - \ln P(-w) = w \quad : \quad \text{slope 1, straight, through } 0$$

Differentiating the mirror identity at the origin forces $P'(0)/P(0) = 1/2$ exactly: the 0/0 at $w = 0$ - the ratio of two vanishing arms - is removed by the theorem to the value 1, the slope of the light line. This is the detailed counterpart of Ch.76's removable value 1 (the TUR-tightness of Gaussian work): one number, two ledgers.

2. The Rigid Line on the Control Instrument

The same stiffness round trip $V = \lambda x^2/2$, $1 \rightarrow 2 \rightarrow 1$, $D = \beta = 1$ (Ch.74/75/76), 250,000 Heun SRK2 runs, no feedback. The mirror measured:

pair	range	slope (thru origin)	bins	
0.15-0.8	0.9682 $\pm$ 0.0211	33	(core, central noise)	
0.8-wing	0.996 $\pm$ 0.01	26	(wing: exactly 1)	
central fills (0/0 $\rightarrow$ 1): +0.80, -0.18, +0.48 (consistent with 1, $\pm$ sampling)				

The integral checks of the same data agree to four digits: $\langle e^{-W} \rangle = 1.0003$, the f-identity $\langle W e^{-W} \rangle = -\langle W \rangle$ (0.9998), and the tilted mean under $e^{-W/2}$ is exactly zero (-0.0005). Yet the variance ledger of Ch.76 reads $q = 0.894$ (skew +1.30, kurtosis 7.0): the distribution is maximally curved and the mirror is perfectly straight. The detailed theorem fixes every RATIO $P(w)/P(-w)$; it says nothing about the SHAPE of P, and the shape is what pays variance. The slack of Ch.76 is real, and it lives strictly off the diagonal.

3. The Coin Leans the Mirror

Feedback breaks the plain theorem exactly as it broke the plain ledger. The far/near bit of Ch.74/75 reads the work and steers the return speed:

row	$\langle W \rangle$	$q(\text{Ch.76})$	mirror slope	$\langle e^{-W} \rangle = J$	net identity
control	+0.1135	0.894	+0.97 ($\rightarrow 1$)	1.0003	fid 1.000
engaged	0.35/2 +0.0432	0.338	-0.25 $\pm$ 0.09	1.0749	fid 4.65 (breaks)
harvest	0.05/16 -0.0392	0 ($W < 0$)	-5.59 $\pm$ 0.25	1.1384	fid -6.43 (breaks)

The engaged coin tilts the mirror by a quarter radian - the excess likelihood of a work drop over a rise now carries the information of the bit - and the harvest inverts it to -5.6. The measured $\langle e^{-W} \rangle$ equals $J_{\text{act}} = 1.075$ and 1.138 of Ch.74/75 to the digit: information moves the variance ledger and the detailed ledger together, one book, one coin (Sagawa-Ueda 2010; Parrondo-Horowitz-Sagawa 2015).

4. The Tangent Has No Mirror

The dragged Gaussian trap ($\mu = +5.98$, $\sigma = 3.47$, $q = 0.9926$) centers 1.7 sigma from the origin: its negative arm carries fewer than one count per bin at the mirror floor, so the detailed ledger is unreadable at this resolution - the tangent is one-armed. Its only readings are the area (the integral ledger, $\langle e^{-W} \rangle = 0.886$, consistent with 1 within its enormous rare-tail error 0.86) and the tight variance ($\text{Var} = 2\langle W \rangle$ to 0.7 CE, Ch.76). Where a Gaussian sits far from zero, even the detailed theorem has nothing to see: the mirror needs both arms, and the tangent borrows its news from the other side that never comes.

5. The 0/0 at $W = 0$, Removed to 1

The ratio $P(W)/P(-W)$ at $W = 0$ is formally 0/0 for a continuous density, and on finite data it is an empty-window estimate. The theorem fills it: nearby, $\ln[P(w)/P(-w)]/w \rightarrow 1$, $P'(0)/P(0) = 1/2$, and the central fills +0.80/-0.18/+0.48 hover at 1 within central sampling noise on the control. This is the same removable value by which Ch.76 priced precision (the TUR edge, $q = 1$ for Gaussian work) and Ch.71 balanced the ledger: the tangent of the light line, the price of knowing per dissipation, and the tight ratio of the parabola are ONE

number, 1. Two ledgers, one book; one removable value, held exactly by the mirror and approached by the variance.

Main Results

THEOREM

The Mirror Is Rigid: on the $\Delta F = 0$, time-reversible control round trip, the detailed ledger $P(w)/P(-w) = e^w$ is exactly the straight line slope 1 (measured 0.996 ± 0.01 in the far wing), the central 0/0 refills to removable value 1, and $\langle e^{-W} \rangle = 1.0003$, $\langle W e^{-W} \rangle = -\langle W \rangle$, $\langle W \rangle_{\text{tilted}} = 0$ - while the variance ledger reads $q = 0.894$. The TUR slack of Ch.76 lives entirely off the mirror, in the shape of P .

COROLLARY

The Coin Leans Both Ledgers Together: engaged feedback tilts the mirror to -0.25 and harvest inverts it to -5.6 , with $\langle e^{-W} \rangle = 1.0749 / 1.1384$ - exactly J_{act} of Ch.74/75. Information is a single entry that debits the variance book and angles the detailed mirror at once.

COROLLARY

The Tangent Is One-Armed: a Gaussian sitting 1.7 sigma from the origin has no negative arm at finite resolution; the detailed ledger goes silent and only the integral form (Jarzynski, within rare-tail error) and the tight variance (Ch.76) remain. The mirror needs both arms; the tangent needs only 1.

Conclusion

Ch.76 priced knowing in variance; Ch.77 opened the detailed ledger beneath it and found it rigid: slope 1 through the origin while the variance book bends. The 0/0 at the mirror's center is filled by the fluctuation theorem to the framework's cardinal removable value, 1 - the same value by which the parabola saturates the TUR, by which the integral ledger balances, and by which reading is priced per dissipation. The demon buys knowing TWICE over the same counter: once in the variance of the work (Ch.76), once in the lean of its mirror (this chapter), and the coin's bill (Ch.75) closes all three accounts at par.

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